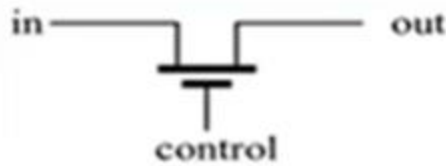


Contents of the lecture

- (1) Switch Primitives in Verilog
- (2) “nmos” Switch
- (3) “pmos” Switch
- (4) Example-1: CMOS Inverter
- (5) Example-2: CMOS 2 input NAND Gate
- (6) Example-2: CMOS 2 input NAND Gate



- (i) Ideal mos switches: nmos, pmos, cmos
- (ii) Resistive mos switches: rnmos, rpmos,rcmos
- (iii) Ideal bidirectional switches: tran, tranif0, tranif1
- (iv) Resistive bidirectional switches: rtran, rtranif0, rtranif1
- (v) Power and Ground nets: supply1, supply0
- (vi) Pullup and Pulldown: pullup, pulldown



nmos		control			
		0	1	x	z
in	0	z	0	L	L
	1	z	1	H	H
	x	z	x	x	x
	z	z	z	z	z

(a) nMOS switch

(i) Declared with keyword: nmos

(ii) Format for instantiation:

nmos [instance name] (output, input, control);
(Drain, Source, Gate)



pmos		control			
		0	1	x	z
in	0	0	z	L	L
	1	1	z	H	H
	x	x	z	x	x
	z	z	z	z	z

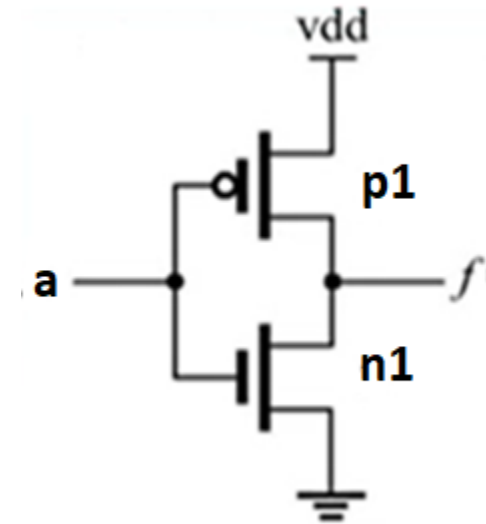
(b) pMOS switch

(i) Declared with keyword: pmos

(ii) Format for instantiation:

pmos [instance name] (output, input, control);

```
module not_my(a,f);  
input a;  
output f;  
supply1 vdd;  
supply0 gnd;  
pmos p1(f,vdd,a);  
nmos n1(f,gnd,a);  
endmodule
```



(i) Format for instantiation:

```
nmos (pmos) [instance name] (output, input, control);  
                                (Drain, Source, Gate)
```